

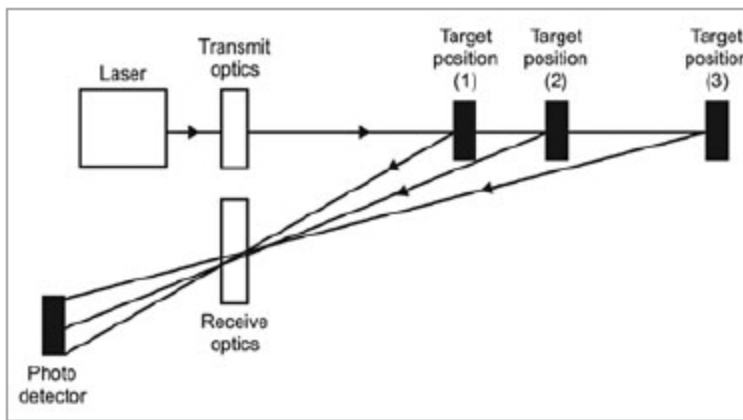


Fig. 5: Basic principle of operation of laser triangulation proximity sensor

functions. In military parlance, the most common application of a proximity sensor is in a laser proximity fuse where command signal is used to initiate detonation of warheads of large artillery shells, aviation bombs and guided missiles.

A number of laser-based range-finding techniques have been exploited successfully for building military rangefinders. These include time-of-flight, phase shift, and frequency-modulated continuous-wave (FM-CW) and triangulation methods of range finding. Suitability of each technique for a given application depends on targeted range, desired range accuracy, range update rate and so on.

In time-of-flight technique, a narrow pulse-width laser beam is transmitted towards the intended target. Target range or distance to target (d) is given by:

$$d = (c \times \Delta t) / 2$$

where d = target range (m), c = speed of light = 3×10^8 m/s and Δt = time interval between transmitted and received laser pulses.

Range accuracy in this case depends on receiver processing speed, and rise and fall times of laser pulses. Range is measured as the time interval between rising or falling edges of the transmitted pulse and the corresponding edge of the received pulse. Uncertainty due to finite values of rise or fall times causes range inaccuracy. Received pulse is processed in a high-speed counter, and clock speed determines processing speed.

Time interval and, hence, range-to-target is measured in terms of number of clock pulses counted by the time interval counter, that is, started by the start pulse corresponding to the leading or trailing edge of the transmitted

laser pulse and stopped by a stop pulse corresponding to the leading or trailing edge of the relevant received pulse. In terms of number of clock cycles, distance-to-target is given by:

$$d = (c \times N) / (2 \times f_{CLK})$$

Here, N is the number of clock pulses counted between start and stop signals, and f_{CLK} is clock frequency. Range inaccuracy in this case is

given by $\pm c/2f_{CLK}$

Worst-case range inaccuracy equals one clock period. Range resolution is determined by laser pulse width, which implies that the laser rangefinder based on time-of-flight principle cannot discriminate between two targets separated in radial range by a distance corresponding to the distance travelled by light in a time period equal to pulse width.

In phase-shift technique of range finding, the laser beam with sinusoidal power modulation is transmitted towards the target, and the diffused or specular reflection from the target is received. The phase of received laser beam is measured and compared with that of the transmitted laser beam. Phase shift is 2π times the product of time-of-flight and modulation frequency. This allows computation of time-of-flight and, hence, distance-to-target from known values of phase-shift and modulation frequency.

Higher modulation frequencies can result in a higher spatial resolution. Phase-shift method appears similar to time-of-flight, as phase shift is proportional to time-of-flight.

FM-CW laser range-finding technique is similar to the one followed in the case of its radar counterpart, that is, FM-CW radar. In this, frequency of a narrow line width laser is modulated with a ramp or sinusoidal signal, collimated and then transmitted towards the target. Received signal corresponding to the reflected laser beam, specular or diffused, is mixed with the reference signal representing the transmitted laser beam. Beat frequency produced as a result of homodyne detection is used to derive range information.

Distance-to-target (d) is given by:

$$d = (f_B \times c \times T_R) / (4 \times \Delta f)$$

where f_B is beat frequency, T_R is ramp waveform time period and Δf is modulation frequency bandwidth.

In triangulation technique, range is determined by using simple laws of trigonometry. Fig. 5 shows the basic principle of operation of a laser triangulation proximity sensor. The sensor has a transmitting channel usually configured around a semiconductor diode laser or a passively Q-switched diode pumped solid-state microchip laser along with associated transmit optics and a receiving channel comprising receiving optics, PIN or APD sensor, and range-processing circuitry.

As shown in the figure, transmitted laser beam reflected from the target located at three different distances produces beam images that are displaced across the active area of the photo sensor. The receiver is designed to produce the focused laser beam spot at the centre of the active area for preset value of proximity distance.

There will be a distance both longer and shorter than the desired distance where the focused beam spot falls just outside the active area. Distance-to-target (d) from the laser transmitter location can be computed from known values of length (l) between transmitter and receiver and angles, (α) and (β), using the following equation:

$$d = (l \times \sin \beta) / \sin(\alpha + \beta)$$

Here, α is the angle made by line-joining transmitter and receiver and line-joining transmitter and target, and β is the angle made by line-joining transmitter and receiver and line-joining receiver and target.

This forms the basis of operation of a laser proximity sensor in general and laser proximity fuse in particular. Performance of a laser proximity sensor can be enhanced by using an axially-symmetric arrangement of multiple aperture photo sensors, as shown in Fig. 6. The design offers better ballistics due to centre of gravity being located on the longitudinal axis of the ammunition round and higher signal-to-noise ratio (SNR) due to averaging of multiple return signals.

Theoretically, all these range-finding techniques, namely, time-of-flight, triangulation, phase-shift and FM-CW, can be used to build laser proximity